Autocorrelation: review the peak rule

One EEG trace and one ACF curve per example

Question to answer

Does the orange point mark the first clear recurrence of the waveform?

Note examples where a small bump is selected, or a convincing earlier recurrence is missed.

The rule being tested

1. Shift the EEG in time and calculate its similarity to itself.
2. Require the ACF to reach zero or below before considering a peak.
3. Select the first positive local maximum within 0.25-2 seconds.

Orange dot: selected peak | **Dashed line: first zero crossing** | **Shading: search interval**

Peak lag = repetition interval. Peak height = strength of repetition.

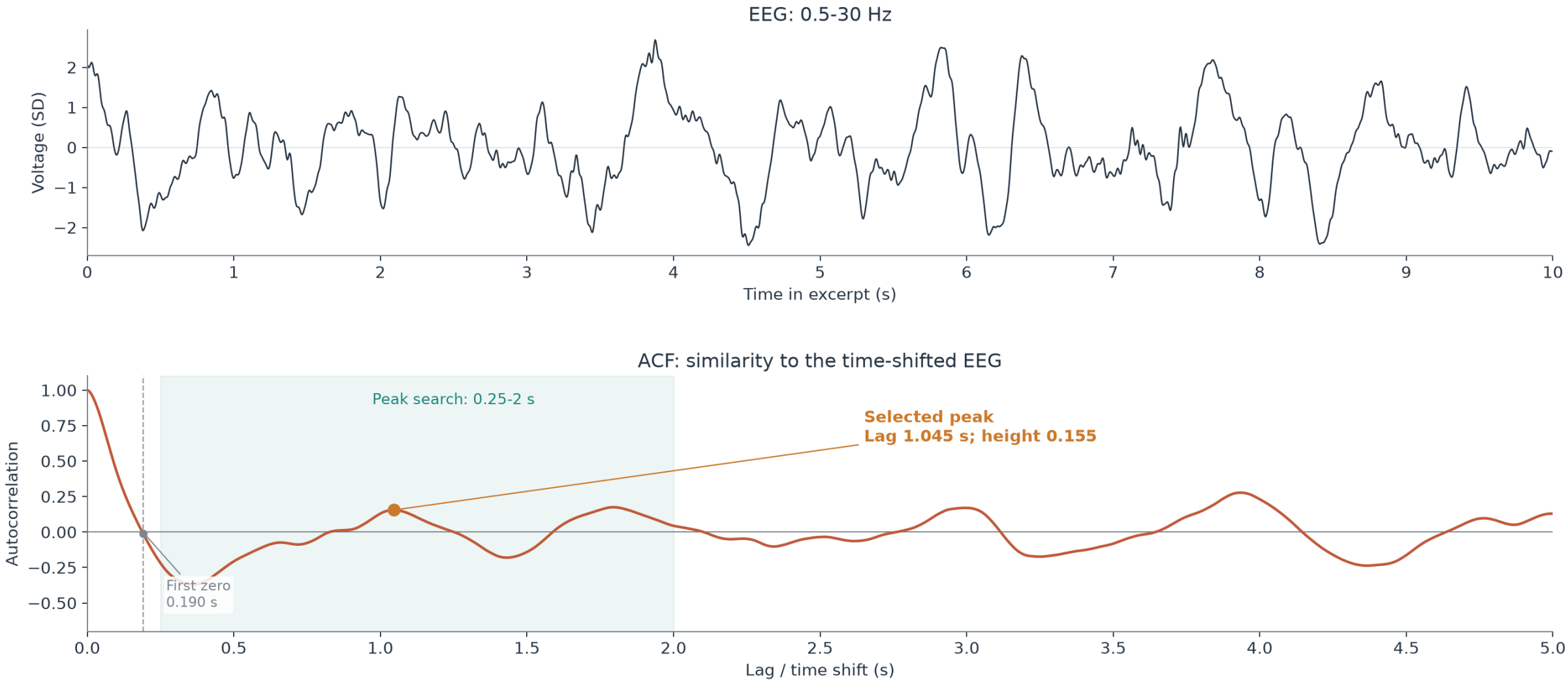
The entire ACF remains visible from 0 to 5 seconds; ACF at zero is 1.

EEG: common average, 0.5-30 Hz, displayed in its own voltage SD. ACF uses the same signal.

The saved ACF and peak measurements are unchanged. The 12 examples match the earlier review.

Autocorrelation peak review

NESI-01 | NESI GRDA >=0.95 | Fz-CAR | 263.000-273.000 s

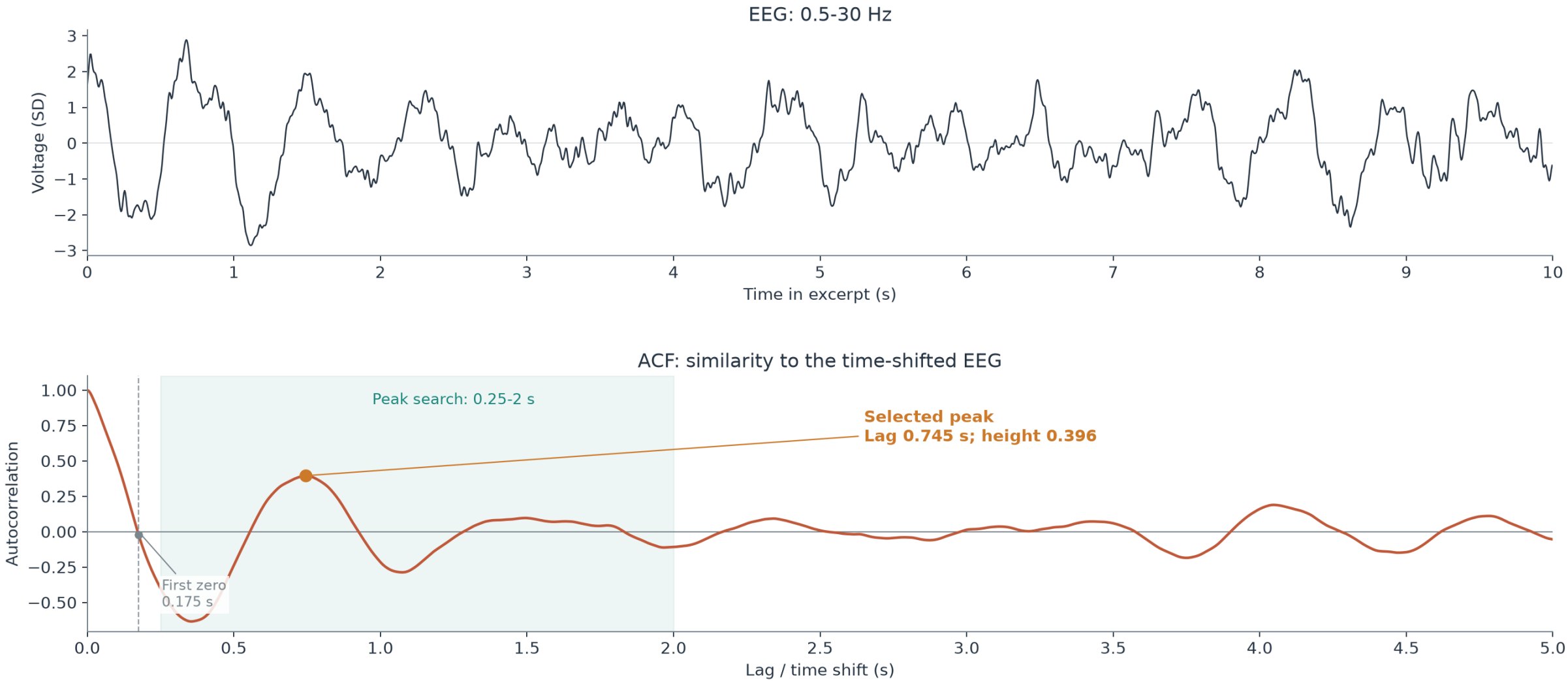


Review: Does the orange point capture the first clear recurrence visible in the EEG?

If a small bump was chosen, or a convincing earlier recurrence was missed, note it.

Autocorrelation peak review

NESI-02 | NESI GRDA >=0.95 | Fp1-CAR | 129.000-139.000 s

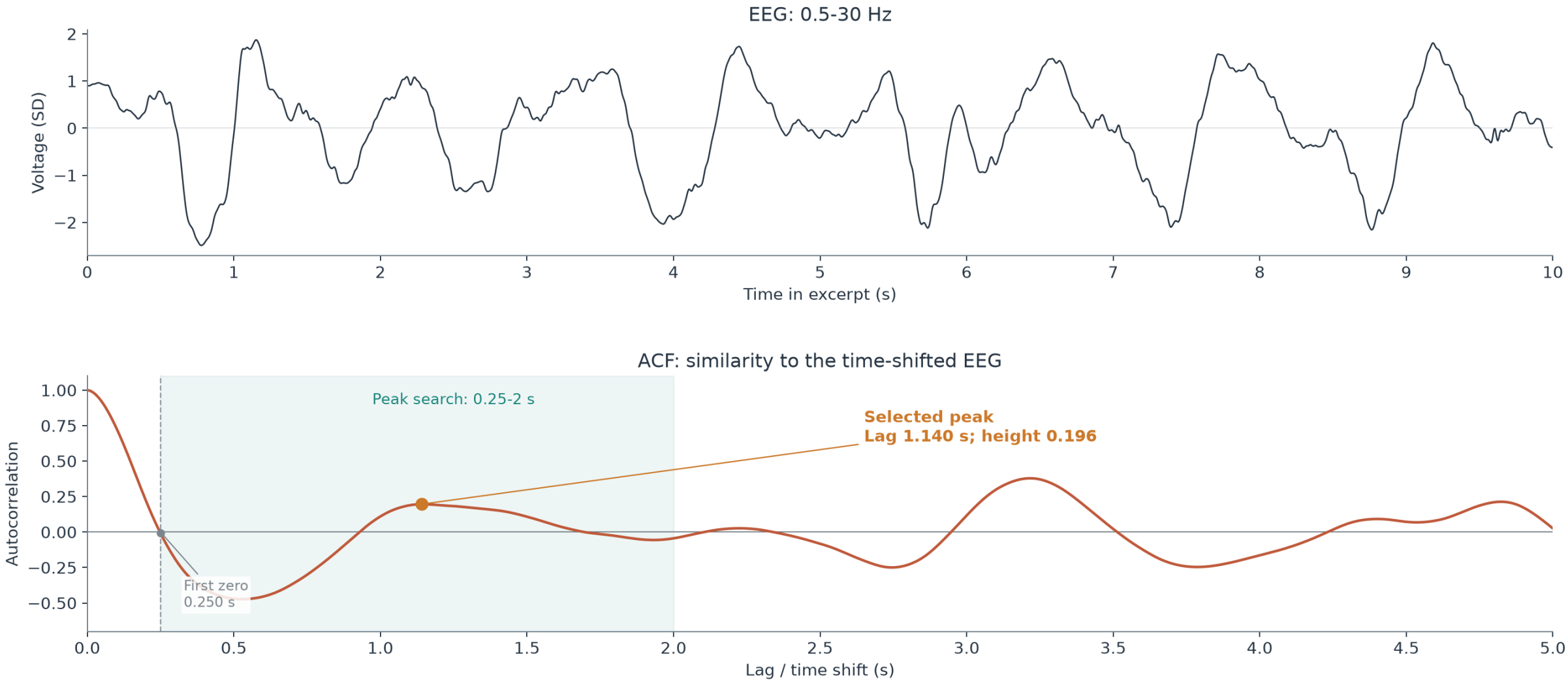


Review: Does the orange point capture the first clear recurrence visible in the EEG?

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Autocorrelation peak review

NESI-03 | NESI GRDA >=0.95 | F7-CAR | 227.000-237.000 s

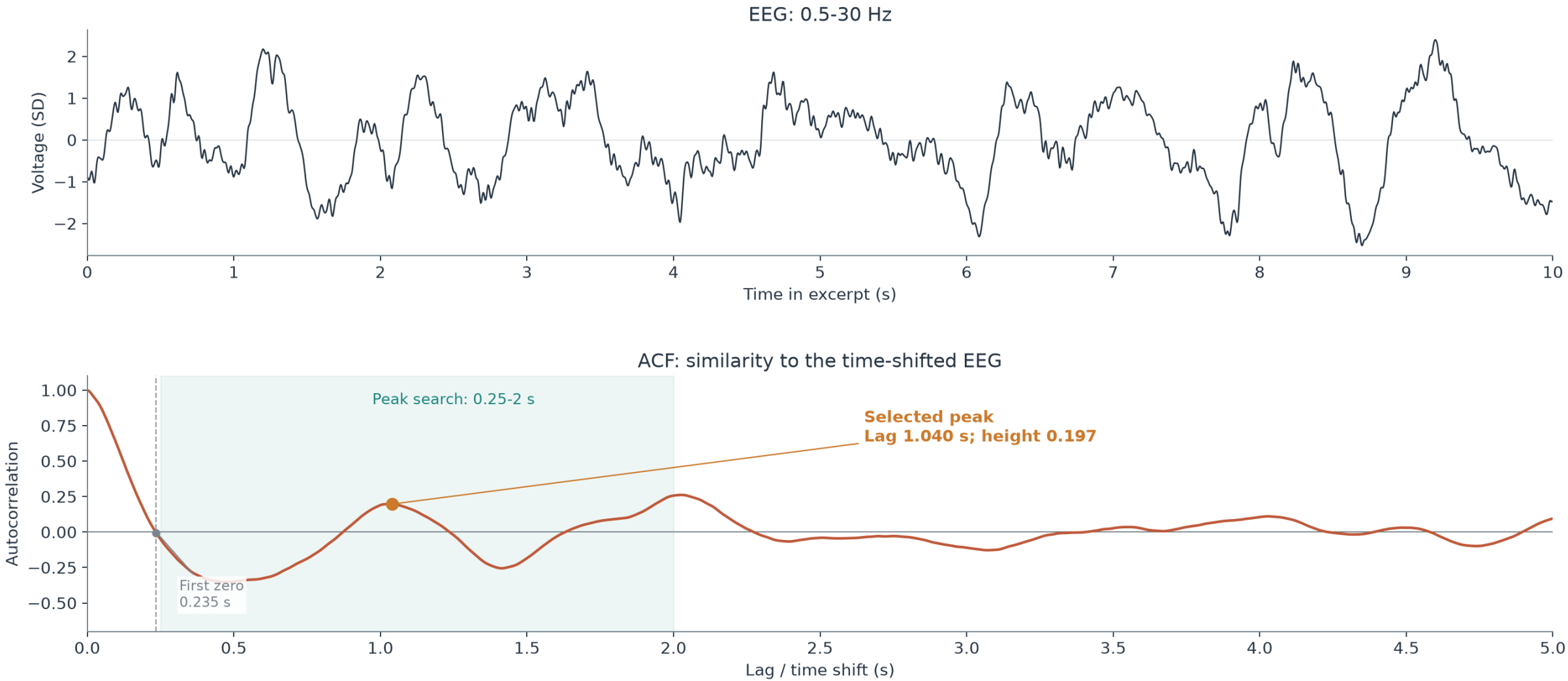


Review: Does the orange point capture the first clear recurrence visible in the EEG?

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Autocorrelation peak review

NESI-04 | NESI GRDA >=0.95 | C3-CAR | 134.000-144.000 s

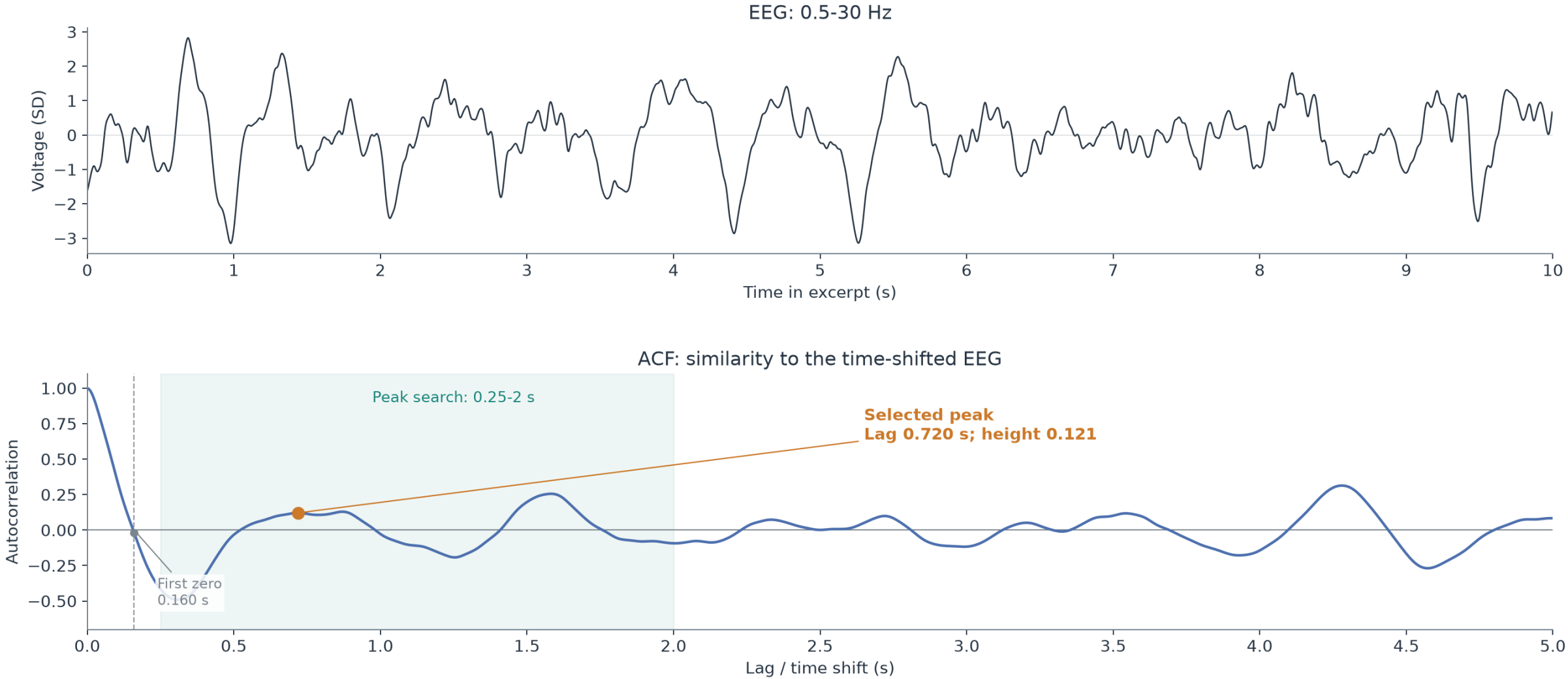


Review: Does the orange point capture the first clear recurrence visible in the EEG?

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Autocorrelation peak review

MASS-01 | MASS N3 | Fz-CAR | 17123.000-17133.000 s

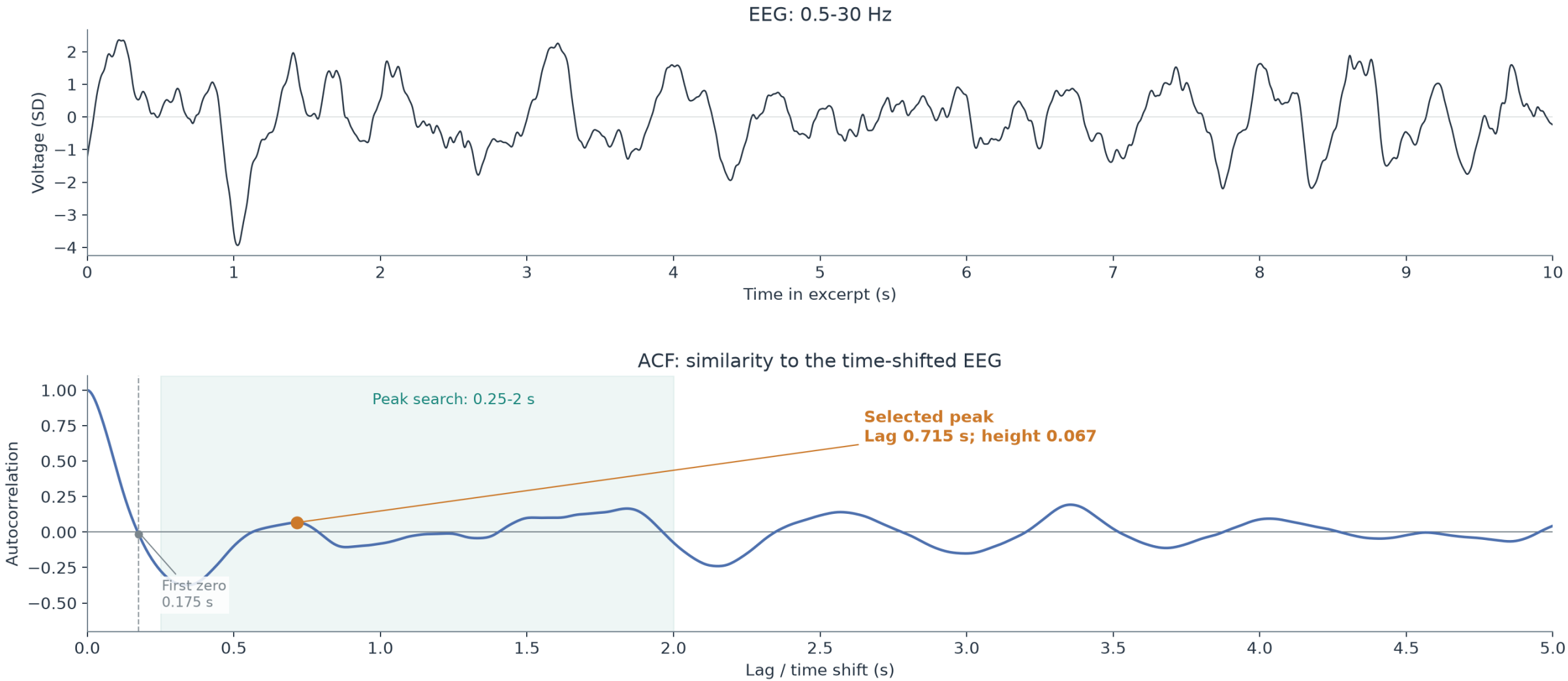


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Autocorrelation peak review

MASS-02 | MASS N3 | Fp1-CAR | 3029.000-3039.000 s

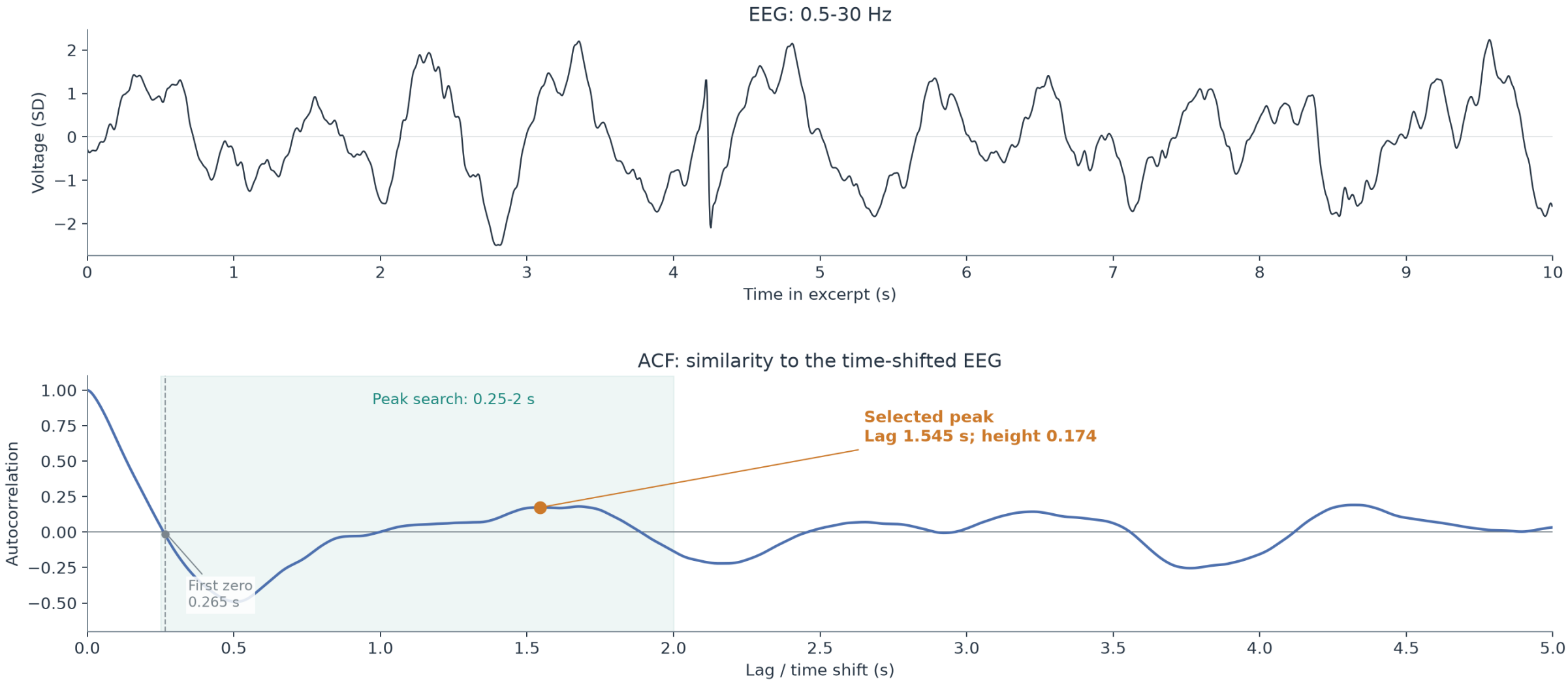


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Autocorrelation peak review

MASS-03 | MASS N3 | F7-CAR | 5153.000-5163.000 s

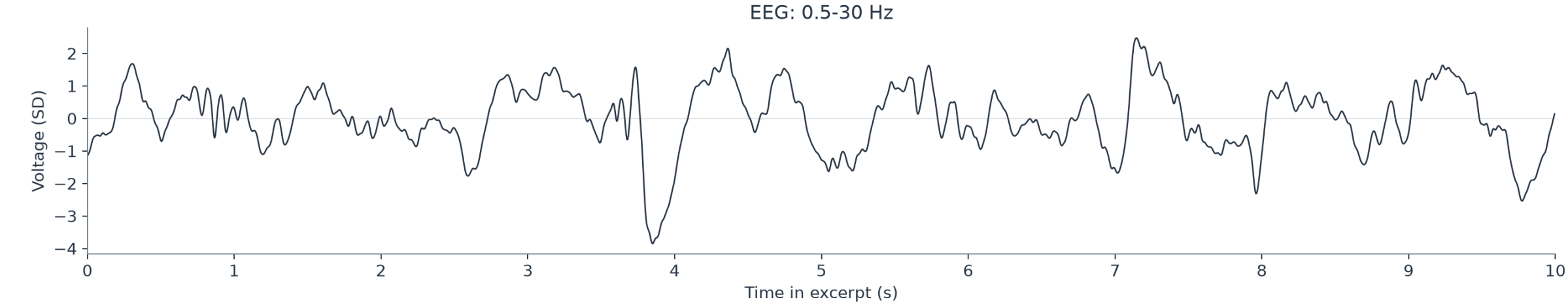


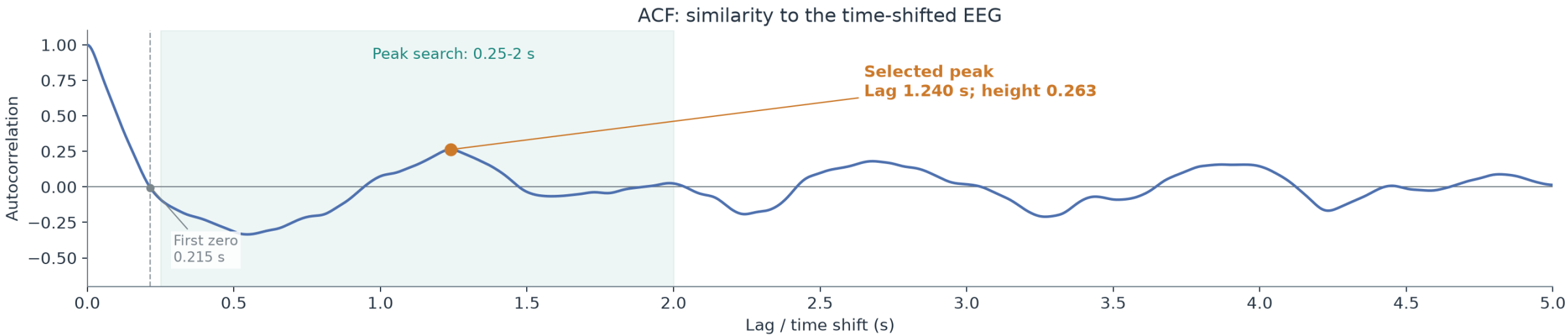
Review: Does the orange point capture the first clear recurrence visible in the EEG?

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Autocorrelation peak review

MASS-04 | MASS N3 | C3-CAR | 10936.000-10946.000 s



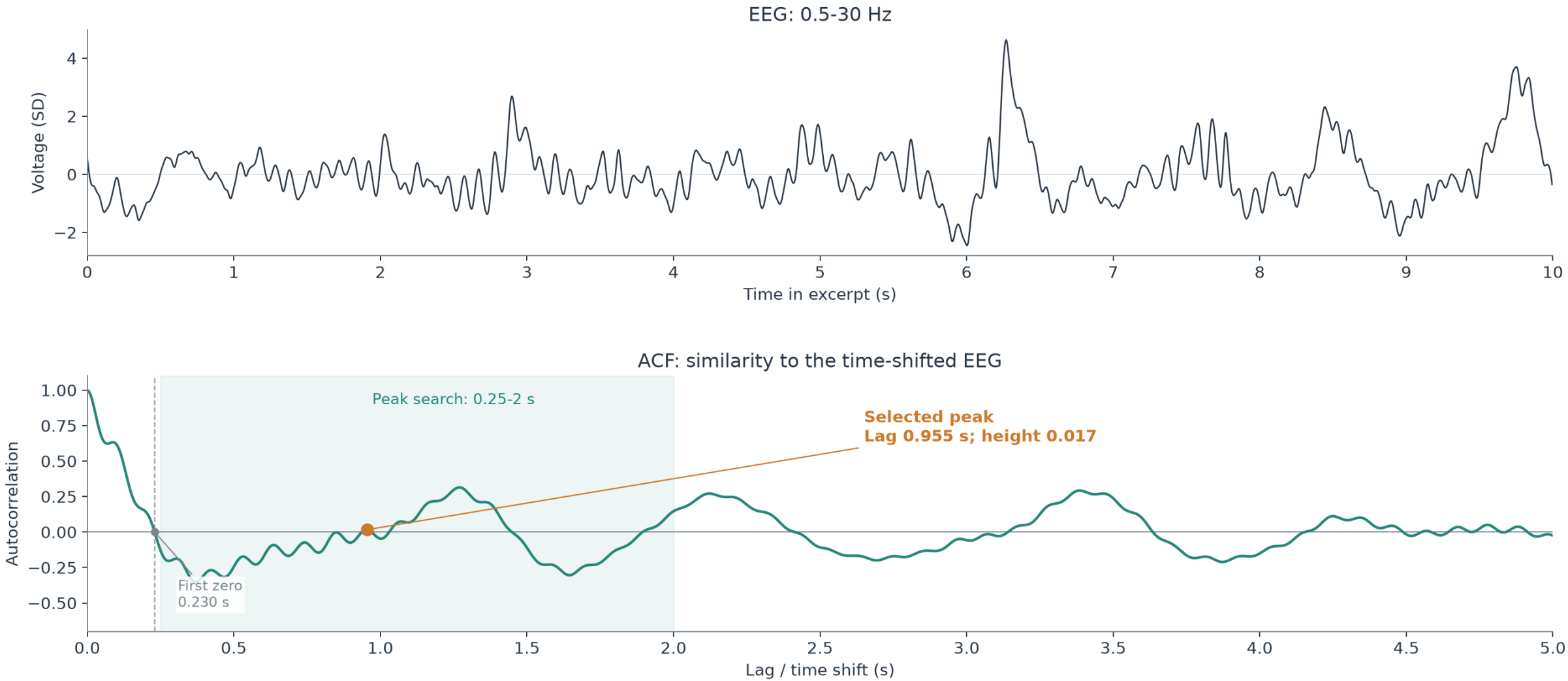


Review: Does the orange point capture the first clear recurrence visible in the EEG?

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Autocorrelation peak review

LENS-01 | LENS N3 | Fz-CAR | 30945.000-30955.000 s

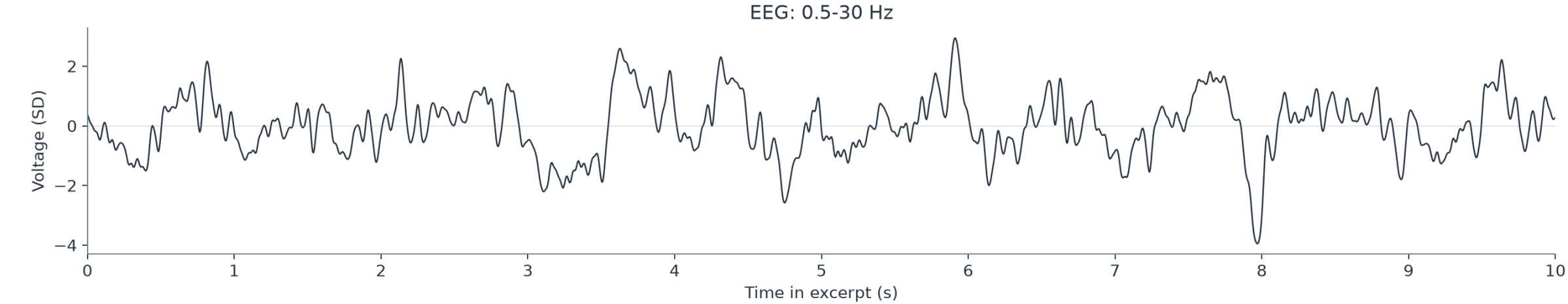


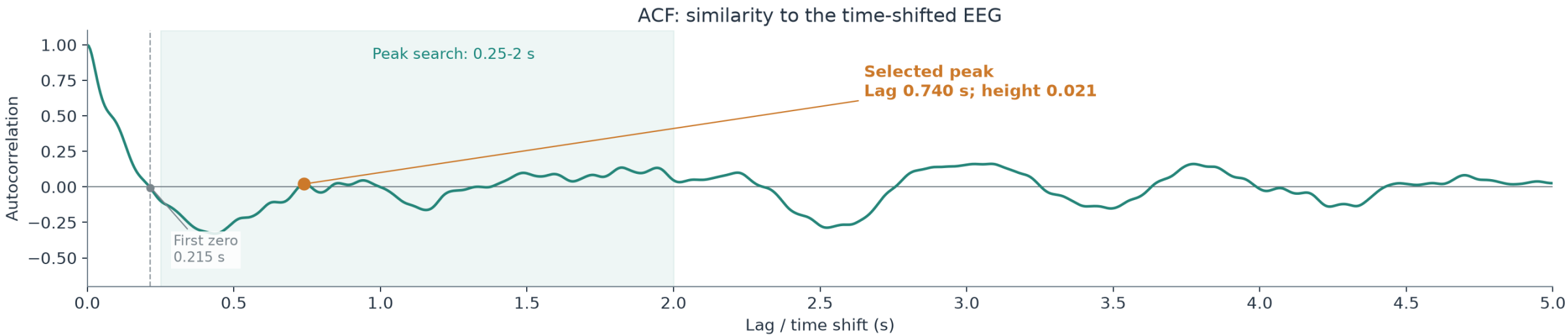
Review: Does the orange point capture the first clear recurrence visible in the EEG?

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Autocorrelation peak review

LENS-02 | LENS N3 | Fp1-CAR | 68355.000-68365.000 s



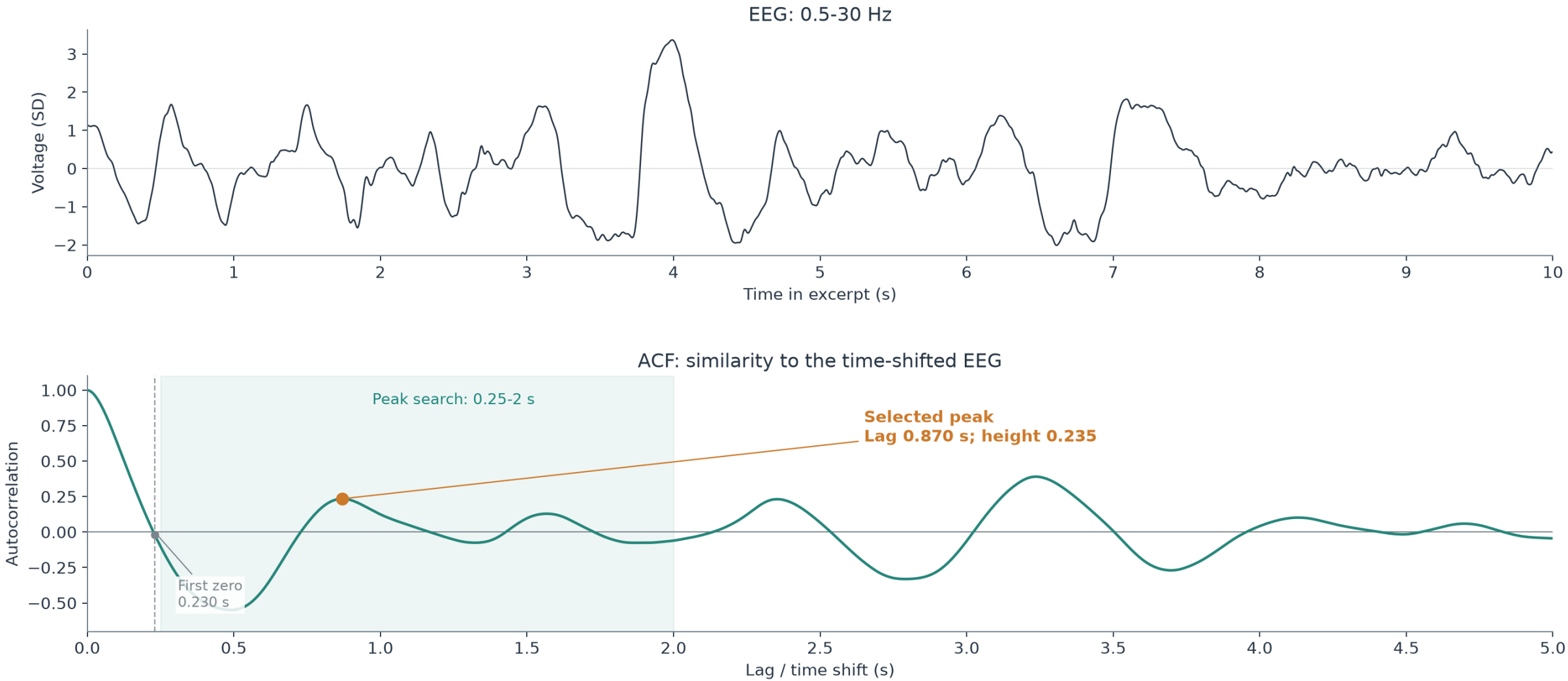


Review: Does the orange point capture the first clear recurrence visible in the EEG?

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Autocorrelation peak review

LENS-03 | LENS N3 | F7-CAR | 41230.000-41240.000 s

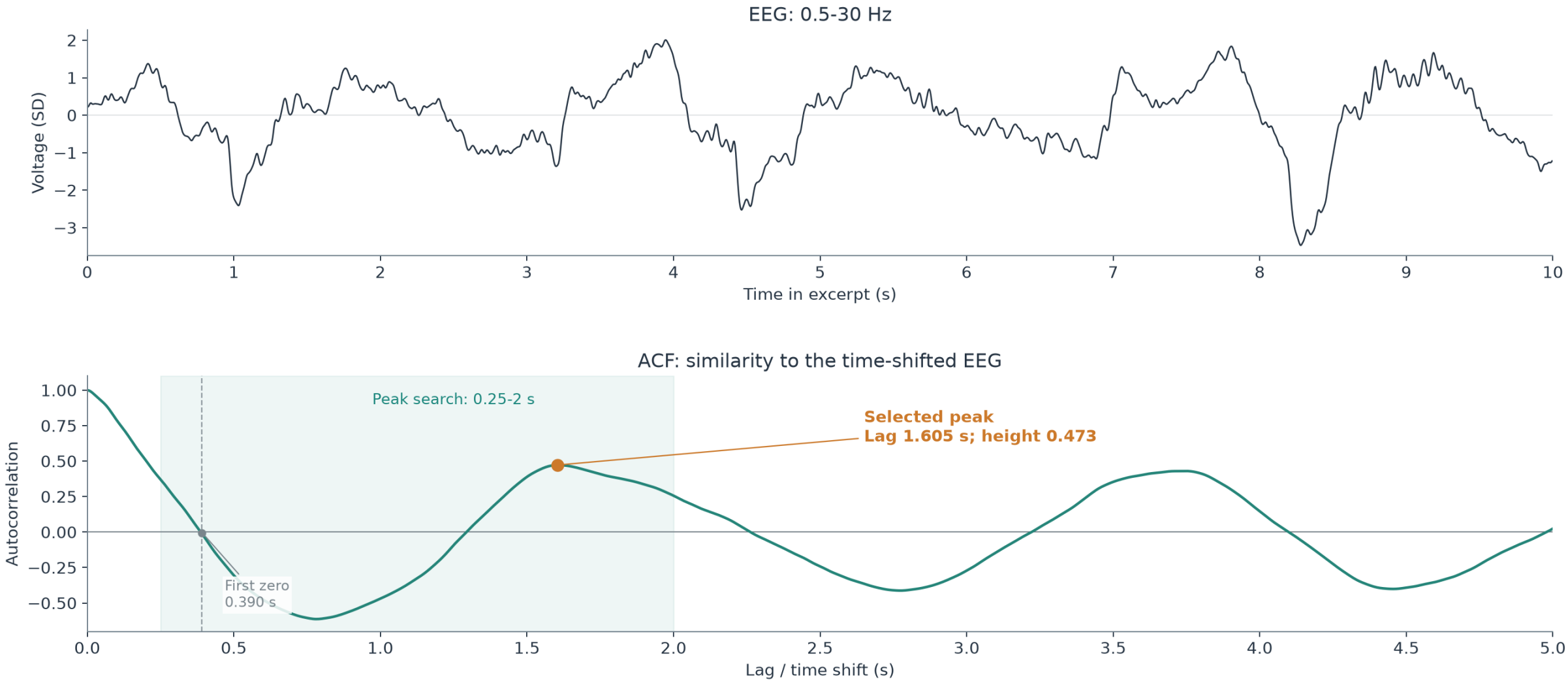


Review: Does the orange point capture the first clear recurrence visible in the EEG?

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Autocorrelation peak review

LENS-04 | LENS N3 | C3-CAR | 55065.000-55075.000 s



Review: Does the orange point capture the first clear recurrence visible in the EEG?

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